Project Report

Track 3: CLI project

Track name: CLI-Based Banking System

Project members:

1.Arijeet Talukder ID:23-51751-2

2.Md Abdur Rahman Shahed ID:23-51741-2

Setup and Run Instructions:

1. Install Python (3.10 or higher)

4. Create Virtual Environment python -m venv .venv

5. Activate Virtual Environment Windows: .venv\Scripts\activate

6. Install Requirements: pip install -r requirements.txt

7. Run python main.py from terminal

3. Features of the System

- Create a new bank account
- Deposit money into an account
- Withdraw money from an account
- Transfer money between accounts
- Search for accounts by name or account number
- Delete an account
- View all accounts
- Calculate total balance
- Display top accounts based on balance

4. System Design

4.1 Modules

The project is divided into multiple modules:

models.py
Contains the Account class which represents a bank account.

manager.py
Handles all business logic such as deposit, withdraw, and transfer.

cli.py
Provides the user interface and menu system using match-case.

main.py
Entry point of the program.

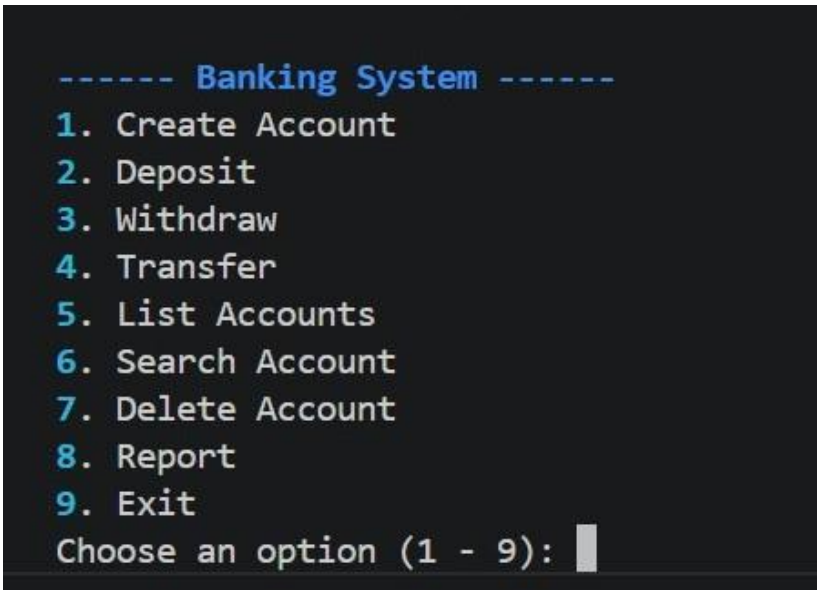
storage.py
Handles saving and loading data using CSV files.

5.Responsibilites:

Arijeet Talukder – main.py, cli.py, storage.py

Md Abdur Rahman Shahed – models.py, manager.py

Screenshots:



```
------ Banking System ------
1. Create Account
2. Deposit
3. Withdraw
4. Transfer
5. List Accounts
6. Search Account
7. Delete Account
8. Report
9. Exit
Choose an option (1 - 9): 1
Enter name: Arijit
Enter initial balance: 343324
Account created! Acc No: 3

------ Banking System ------
1. Create Account
2. Deposit
3. Withdraw
4. Transfer
5. List Accounts
6. Search Account
7. Delete Account
8. Report
9. Exit
Choose an option (1 - 9): 2
Account number: 1
Amount: 10000
Deposit successful
```

```
------ Banking System ------
1. Create Account
2. Deposit
3. Withdraw
4. Transfer
5. List Accounts
6. Search Account
7. Delete Account
8. Report
9. Exit
Choose an option (1 - 9): 3
Account number: 2
Amount: 2010
Withdraw successful

------ Banking System ------
1. Create Account
2. Deposit
3. Withdraw
4. Transfer
5. List Accounts
6. Search Account
7. Delete Account
8. Report
9. Exit
Choose an option (1 - 9): 4
From Account: 3
To Account: 1
Amount: 1000
Transfer successful
```

```
------ Banking System ------
1. Create Account
2. Deposit
3. Withdraw
4. Transfer
5. List Accounts
6. Search Account
7. Delete Account
8. Report
9. Exit
Choose an option (1 - 9): 5
```

Accounts

Acc No	Name	Balance
1	shahed	403002.00
2	Rahman	1424.00
3	Arijit	342324.00

```
------ Banking System ------
1. Create Account
2. Deposit
3. Withdraw
4. Transfer
5. List Accounts
6. Search Account
7. Delete Account
8. Report
9. Exit
Choose an option (1 - 9): 6
Enter name or account number: rahman
```

```
------ Banking System ------
1. Create Account
2. Deposit
3. Withdraw
4. Transfer
5. List Accounts
6. Search Account
7. Delete Account
8. Report
9. Exit
Choose an option (1 - 9): 6
Enter name or account number: rahman
```

Search Results

Acc No	Name	Balance
2	Rahman	1424.00

```
------ Banking System ------
1. Create Account
2. Deposit
3. Withdraw
4. Transfer
5. List Accounts
6. Search Account
7. Delete Account
8. Report
9. Exit
Choose an option (1 - 9): 7
Account number: 3
Account deleted
```

```
5. List Accounts
6. Search Account
7. Delete Account
8. Report
9. Exit
Choose an option (1 - 9): 8

--- REPORT ---
Total Balance: 404426.0
```

Top Accounts

Acc No	Name	Balance
1	shahed	403002.00
2	Rahman	1424.00

```
------ Banking System ------
1. Create Account
2. Deposit
3. Withdraw
4. Transfer
5. List Accounts
6. Search Account
7. Delete Account
8. Report
9. Exit
Choose an option (1 - 9): 9
Exiting... Goodbye!
(.venv) PS C:\Users\abut1\OneDrive\Documents\New folder>
```